

Block-Wise Latent Genotypes Revisited: A Hierarchical Spike-and-Slab Framework for GWAS Fine-Mapping, Generalized and Benchmarked on Ten 1000 Genomes Loci

Ian Johnston¹

¹Independent researcher

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Abstract

Genome-wide association studies must select causal variants from among many correlated, LD-linked SNPs – a problem Johnston [2015]’s Chapter 4 addressed by pre-processing each linkage-disequilibrium block into one decorrelated latent genotype via a simultaneous-autoregressive model, fit by hand-derived Newton approximations and validated only on two case studies, never against a dedicated fine-mapping method. We reimplement this model with modern autodiff (JAX in place of hand-derived Taylor expansions), fix two real numerical bugs surfaced only by running it on real chromosomal coordinates for the first time, and package it as a reusable, sklearn-shaped Python library. We then close the missing-benchmark gap directly: ten real 1000 Genomes Phase 3 loci – from textbook single- population selective sweeps to disease-associated variants of more moderate effect – each evaluated against standard classifiers (L1-logistic, PCA, Random Forest, linear SVM) and against SuSiE [Wang et al., 2020], the current standard for Bayesian genetic fine-mapping. The result is a mixed, honestly-reported picture rather than a clean win: block-latent decorrelation sometimes localizes the literature-established causal SNP more precisely than SuSiE and sometimes loses to it outright, and which happens co-varies with how saturated the underlying population-differentiation signal is. We characterize this pattern across all ten loci and report it as the paper’s central empirical finding. A follow-up hyperparameter sensitivity study across the same ten loci finds a further honest complication: hierboost’s classification accuracy is remarkably insensitive to its own tuning parameters, but its localization precision is highly and non-monotonically sensitive to them – and CV accuracy, the only tuning signal available without already knowing the causal SNP, is an unreliable guide to which setting actually improves localization, sometimes helping and sometimes actively misleading. Finally, we test whether combining hierboost, SuSiE, and a classical single-SNP test into a rank-averaging ensemble beats any one method alone: it does, modestly (lowest mean and median localization distance of the four, beating hierboost alone by $\sim 11\%$), but not uniformly – one locus shows the ensemble actively *outvoting* a correct minority answer, an honest limitation rather than a clean win.

1 Introduction

Genome-wide association studies face a large- p -small- n variable selection problem made harder by linkage disequilibrium (LD): SNPs near each other on a chromosome are non-randomly correlated, so a single true causal signal is typically tagged by many redundant, highly-correlated markers. Naive per-SNP association testing or standard spike-and-slab variable selection spreads posterior

inclusion probability across all of them, diluting statistical power and making the “which SNP is actually causal” question – *fine-mapping* – hard to answer directly. Johnston [2015]’s Chapter 4 addressed this by pre-processing each LD block into one decorrelated latent genotype (Eq. 3) before a spike-and-slab GLM ever sees the data, fit via a hand-derived Newton approximation and validated only on WTCCC/GAW18 case studies without benchmarking against a dedicated fine-mapping method.

In the decade since, fine-mapping has matured into its own subfield: SuSiE [Wang et al., 2020], FINEMAP [Benner et al., 2016], and their descendants are now the default tools for exactly this problem, and none existed when Chapter 4 was written. This paper (1) re-implements Chapter 4’s model with modern autodiff (JAX) in place of hand-derived Taylor approximations, fixing two real numerical bugs surfaced only by running it on real chromosomal coordinates for the first time (Sec. 3.2); (2) packages it as a reusable, sklearn-shaped Python library (**hierboost**); and (3) closes the missing-benchmark gap directly: ten real 1000 Genomes Phase 3 loci, spanning textbook single-population sweeps (EDAR, ABCC11, DARC/ACKR1) to disease-associated variants of more moderate effect (APOL1, FUT2), each benchmarked against standard classifiers *and* against SuSiE.

The result is not a clean win for either method. Sec. 4 and 6 report the honest, mixed picture: the block-latent approach is sometimes competitive with or exceeds SuSiE’s localization accuracy, sometimes loses to it outright, and the direction of that difference co-varies with how saturated/extreme the underlying population-differentiation signal is – a genuine empirical finding, not a clean methods-paper narrative, and reported as such.

Contributions.

- A modernized, open-source reimplementation of the dissertation’s block-wise latent genotype model (JAX autodiff, no hand-derived Taylor approximation), with two real numerical bugs found and fixed running it on real genomic coordinates for the first time (Sec. 3.2).
- The first head-to-head benchmark of this model against SuSiE, the current standard for Bayesian fine-mapping, on ten real, independent 1000 Genomes loci (Sec. 4).
- An honest synthesis of *when* block-latent decorrelation helps relative to both standard classifiers and SuSiE, extending a similar cross-domain synthesis from non-genomic applications of the same underlying machinery (Sec. 6).
- A reusable, sklearn-shaped Python package (**hierboost**) and a reproducible per-locus pipeline (fetch → block → CV → full-fit → JSON summary → auto-generated results table) so this comparison can be extended to new loci without hand-transcribing results.
- A hyperparameter/prior sensitivity study across the same ten loci (Sec. 5) showing accuracy and localization respond very differently to tuning, and that CV accuracy – the only tuning signal available without ground truth – does not reliably track localization quality, an honest complication for anyone hoping to tune this model in practice.
- A ground-truth-free tuning criterion built and tested directly in response to that gap (Sec. 5): a SuSiE-purity-inspired score modestly outperforms CV accuracy at recovering oracle-best localization, with a diagnosed (not merely observed) singleton-block failure mode that a naive fix does not cleanly repair.

2 Related Work

Spike-and-slab variable selection. The mixture prior in Eq. 2 traces to George and McCulloch [1993]; the dissertation’s continuous relaxation and EM-filter fitting strategy is specific to the large- p -small- n , LD-correlated GWAS setting.

Modern Bayesian fine-mapping. SuSiE [Wang et al., 2020] and its summary-statistics variant [Zou et al., 2022] are the current default tool for exactly the problem Chapter 4 targets: identifying which of several tightly LD-linked SNPs is causal. SuSiE’s “Sum of Single Effects” model handles correlated predictors by letting them share a *credible set* rather than pre-averaging them into one shared latent, a structurally different strategy from the block-latent decorrelate-then-select pipeline this paper benchmarks it against (Sec. 3.4). Benner et al. [2016]’s FINEMAP is a comparable shotgun-stochastic-search alternative operating on summary statistics; both post-date the 2015 dissertation this paper builds on and neither was available for comparison at the time.

Genomic block-latent / factor-analytic approaches to LD. Modeling a correlated LD block via a shared underlying factor is closely related to principal-components-based adjustment for population structure [e.g. Price et al., 2006], though those methods target genome-wide stratification rather than local per-block redundancy. Chapter 4’s SAR formulation (Eq. 3) is a direct adaptation of spatial-econometric simultaneous autoregressive models [LeSage and Pace, 2009] to genomic distance as the notion of “space.”

3 Methods

3.1 Spike-and-slab base model

Chapter 2 of Johnston [2015] (“Spatial Boost Model”, later published as Johnston et al. 2016) poses GWAS as Bayesian variable selection on a logistic regression: for n individuals with genotype dosages $X_i \in \{0, 1, 2\}^p$ and binary phenotype y_i ,

$$y_i \mid X_i, \beta \stackrel{\text{ind}}{\sim} \text{Bernoulli}\left(\text{logit}^{-1}(\beta_0 + X_i^\top \beta)\right), \quad (1)$$

with a continuous relaxation of the classical spike-and-slab prior [George and McCulloch, 1993] on each coefficient,

$$\beta_j \mid \theta_j, \sigma^2 \stackrel{\text{ind}}{\sim} \text{Normal}(0, \sigma^2[\theta_j \kappa + (1 - \theta_j)]), \quad j = 1, \dots, p, \quad (2)$$

where $\theta_j \in \{0, 1\}$ and $\kappa > 1$ separates the “spike” (near-zero variance, $\theta_j = 0$) from the “slab” (variance inflated by κ , $\theta_j = 1$) component – a continuous relaxation of exact-zero variable selection chosen specifically because it admits an EM algorithm (Sec. 3.3). Chapter 2’s full model additionally places a logit-linear prior on $\mathbb{P}(\theta_j = 1)$ driven by each SNP’s kernel-weighted proximity to relevance-annotated genes – the “gene boost” that gives the dissertation and this project’s non-genomic generalizations their name. The case studies in Sec. 4 do *not* use this external-annotation boost (there is no independent gene-relevance signal to inject for a pure population-differentiation task); they use Eqs. 1–2 with a flat prior on θ_j downstream of the block-latent decorrelation in Sec. 3.2, i.e. Chapter 4’s contribution in isolation. The boosting-prior mechanism itself is retained in `hierboost` (`kernels.py`, generalized past genes to arbitrary annotated groups) and used elsewhere in this project’s broader generalization work, but is out of scope for the fine-mapping-style comparison this paper focuses on.

3.2 Block-wise latent genotypes

This section recaps Chapter 4 of Johnston [2015] (“Block-Wise Latent Genotypes”), which the case studies below apply directly to real data for the first time.

Block definition. A chromosome is partitioned into non-overlapping blocks by a distance threshold ζ : two adjacent SNPs at positions s_j, s_k fall in the same block iff $|s_j - s_k| \leq \zeta$. The dissertation recommends $\zeta \leq 10^4$ bp as a practical balance between preserving the inverse relationship between genomic distance and LD and keeping the number of blocks computationally tractable; our implementation (`hierboost.structure.determine_blocks`, method “`threshold`”) instead defaults ζ to the 75th percentile of consecutive-gap distances on the observed coordinates, a data-adaptive “natural breaks” choice requiring no locus-specific tuning.

Simultaneous autoregressive (SAR) latent model. For individual i , let $X_i \in \{0, 1, 2\}^p$ be the observed genotype dosages for p SNPs in one block. The model introduces independent latent genotypes $Z_i \sim \text{MVN}(\mu, \Sigma)$ with diagonal $\Sigma = \tau^2 I$, then a spatially correlated latent U_i via the SAR relation

$$U_i = BU_i + Z_i, \quad (3)$$

where B is a matrix of non-negative spatial weights ($B_{jj} = 0$) encoding proximity between SNPs j, k in the block:

$$B_{jk} = \Phi\left(\frac{-|s_j - s_k|}{\phi_j}\right) + \Phi\left(\frac{-|s_j - s_k|}{\phi_k}\right), \quad (4)$$

Φ the standard normal CDF and ϕ_j a per-SNP bandwidth controlling the radius of influence (`hierboost.kernels.sar_weight_matrix`; ϕ auto-fit against $|\text{corr}(X)|$ by `fit_gaussian_bandwidth`, generalizing the dissertation’s Sec. 5.1 LD-decay fit). Writing $C = (I - B)^{-1}$, Eq. 3 induces $U_i \sim \text{MVN}(C\mu, C\Sigma C^\top)$. The observed dosage is then a censored/quantized view of the spatially-smoothed latent:

$$X_{ij} \mid Z_i \stackrel{\text{ind}}{\sim} \text{Binomial}\left(2, \text{logit}^{-1}[C_j^\top Z_i]\right). \quad (5)$$

Fitting Z_i (and the hyperparameters θ, σ^2, γ governing the downstream spike-and-slab prior/outcome model) requires alternating Newton updates on Eq. 5’s non-conjugate binomial-logit likelihood. The dissertation derives these updates by hand (Taylor-expanding the logistic link); our implementation (`hierboost.latent`) instead differentiates the exact log-likelihood with JAX `grad/hessian` and uses reparameterized Monte Carlo for the moment-matching step, removing the hand-derived approximation while keeping the same alternating-Newton structure.

Continuous relaxation. When the outcome/feature model is Gaussian rather than Binomial (every non-genomic application in this project, and the “continuous branch” used for the CV comparisons below), Eq. 5 is replaced by a linear-Gaussian observation model, making the whole block latent a one-factor probabilistic PCA problem with a closed-form solution (`hierboost.factor.gaussian_block_fa` – no Newton iteration at all. Section 4’s CV comparisons use this branch by default, with the literal Binomial branch (Eq. 5) run separately as a full-data comparison where noted, since it does not currently support k-fold CV at the same computational cost (Sec. 3.3).

Two bugs found reproducing this on real data. Porting Eq. 5’s fit to real chromosomal coordinates (rather than the dissertation’s simulated data) surfaced two implementation bugs worth recording for anyone else doing this: (1) coordinates cast to `float32` lose precision at real genomic

bp magnitude ($\sim 1.3 \times 10^8$) – two SNPs 1bp apart collapsed to an identical float32 value, corrupting Eq. 4’s distance matrix; fixed by centering each block’s coordinates on its own minimum before casting (exact, since B only depends on pairwise differences). (2) undamped Newton steps in Eq. 5’s fit occasionally diverge in a runaway feedback loop through the outcome regression; fixed with a max-step clip (`max_step=5.0`). Both fixes are on by default in the code used for every result below.

3.3 Inference

Three inference routes are implemented for Eqs. 1–2 (`hierboost.spike_slab`): an EM algorithm treating θ as missing data (fast, point estimates of $\mathbb{P}(\theta_j=1 \mid y)$ via $\text{logit}^{-1}(S_j)$ where S_j is each SNP’s linear predictor under the fitted prior); a Pólya-Gamma-augmented Gibbs sampler [Polson et al., 2013] giving full posterior samples but at substantially higher cost; and *EM-filter*, which runs the EM step once to rank features, retains a data-driven top fraction via *EMBFDR* (a Bayesian-false-discovery-rate-style automatic threshold on κ , Ch. 2 Sec. 2.2.1), and reports the EM-fitted $\hat{\theta}_j$ on the retained set – the dissertation’s own finding (Ch. 3 Sec. 3.3.1: “I can achieve roughly the same level of performance ... using the final estimates of $\text{logit}^{-1}(S_j)$ after running the EM filter in place of ... the Gibbs sampler”) that filtering is nearly as accurate as full Gibbs sampling at a small fraction of the cost. All CV comparisons in Sec. 4 use EM-filter for exactly this reason: 10-fold CV $\times$ multiple methods $\times$ up to ~ 700 SNPs makes a full Gibbs sampler impractical to run at the scale this paper needed, and the dissertation’s own calibration study already established the accuracy cost of skipping it is small.

The discrete/Binomial block-latent branch (Eq. 5) is fit only once per locus on a single train/test split rather than under k-fold CV, for the compute-cost reason given in Sec. 3.2: JAX recompiles per unique block size, and with 100+ variably-sized LD blocks per locus that cost multiplies badly across folds. This is a real limitation of the current implementation, not a property of the model, and is noted explicitly in Sec. 8.

3.4 SuSiE as a fine-mapping baseline

3.5 Triangulation: combining hierboost, SuSiE, and single-SNP tests

Sec. 6 treats hierboost and SuSiE as competitors. We also test whether combining their per-SNP evidence with a third, much simpler baseline – a classical single-SNP association test (per-SNP Pearson correlation between raw dosage and the ancestry label, the linear- probability-model analogue of a GWAS single-marker test) – outperforms any one method alone. Each method’s evidence is placed on a common per-SNP scale (hierboost: its LD block’s θ , broadcast to every member SNP; SuSiE: posterior inclusion probability; single-SNP: test statistic) and combined by rank-averaging each SNP’s percentile rank across the three methods, then taking the top-ranked SNP as the ensemble’s pick. Two alternative combination rules (a floored geometric mean, and a rule that only triangulates when two methods’ own top picks physically agree within 10kb) were also tested; rank-averaging was the most consistently reliable of the three and is reported as the primary rule in Sec. 7.

3.6 Data

4 Case Studies

Table 1 summarizes all ten loci: eight are textbook examples of strong, well-replicated single-locus selection or trait association with a literature-established causal or lead SNP; the choice deliberately spans selection strength/differentiation (from LCT’s moderate EUR/AFR divergence to DARC/APOL1’s near-fixed extremes) and population contrast (EUR/AFR, EAS/EUR), following the same “does this work, and when” question the cross-domain generalization work already raised for other datasets.

Table 1: All loci run to date. “Best baseline” is the best of L1-logistic/PCA+logistic/linear SVM/Random Forest. Distances are from each method’s top-ranked SNP/block to the literature-reported causal/lead variant.

Locus	Contrast	Best baseline	hierboost	hb dist.	SuSiE	SuSiE dist.
LCT	EUR vs AFR	98.7% (659.0f)	95.4% (17.6f)	38033bp	97.3% (27.0f)	78255bp
SLC24A5	EUR vs AFR	99.4% (28.5f)	99.4% (9.8f)	7010bp	99.3% (2.0f)	149418bp
DARC/ACKR1	EUR vs AFR	99.2% (723.0f)	99.1% (11.3f)	411bp	99.1% (1.0f)	0bp
EDAR	EAS vs EUR	99.1% (28.5f)	98.5% (9.9f)	62579bp	98.2% (29.0f)	26052bp
HERC2/OCA2	EUR vs AFR	98.5% (137.0f)	97.8% (14.6f)	82072bp	97.9% (36.7f)	79616bp
ABCC11	EAS vs EUR	94.4% (62.0f)	91.0% (14.8f)	53453bp	92.4% (71.3f)	18625bp
ADH1B	EAS vs EUR	98.7% (671.0f)	97.0% (14.7f)	6695bp	97.5% (56.6f)	96539bp
APOL1	AFR vs EUR	99.1% (90.3f)	98.5% (18.9f)	5530bp	97.9% (15.3f)	64765bp
FUT2	EUR vs EAS	98.8% (116.0f)	97.9% (17.0f)	74040bp	98.2% (34.1f)	62432bp
SLC45A2	EUR vs AFR	98.5% (120.0f)	97.9% (13.2f)	67057bp	98.2% (7.4f)	0bp

4.1 LCT (lactase persistence)

The textbook example of strong recent positive selection in humans [Bersaglieri et al., 2004]: the causal regulatory variant rs4988235 (chr2:136,608,646, GRCh37) confers lactase persistence into adulthood and is common in European ancestry, rare elsewhere. 1164 individuals, 659 SNPs ($MAF \geq 0.05$), 167 LD blocks, EUR vs AFR. The best baseline is linear SVM at 98.7% using all 659 raw SNPs; hierboost reaches 95.4% CV accuracy with a mean of 17.6 retained features ($37-90 \times$ fewer than the baselines), and SuSiE reaches an intermediate 97.3% with 27 credible-set SNPs. On localization, hierboost’s top full-data block sits 38.0kb from rs4988235 – inside the LCT gene body itself (136,545,410–136,594,750bp), a biologically sensible near-miss – while SuSiE’s top PIP SNP ($pip=1.0000$) lands 78.3kb away, roughly twice as far. The separately-run discrete/Binomial branch (Sec. 3.2; single train/test split, not CV) reaches 86.3% held-out accuracy with its top block only 8.1kb from rs4988235 – closer than either the continuous branch or SuSiE, at a real accuracy cost.

4.2 SLC24A5 (light skin pigmentation)

1164 individuals, 461 SNPs, 117 LD blocks, EUR vs AFR. This locus is the cleanest CV result of the ten: hierboost (99.40%, 9.8 features) ties the best baseline (L1-logistic, 99.40%, 28.5 features) using roughly a third as many features, and SuSiE reaches 99.31% with only 2 credible-set SNPs – all four numbers within noise of each other. Localization tells a different, more interesting story: hierboost’s top block sits 7.0kb from the causal SNP rs1426654 (retained, not dropped, at rank

3/11 by inclusion probability), but SuSiE’s top PIP SNP (pip=1.0000, only 2 credible sets) is 149.4kb away – SuSiE’s near-perfect CV accuracy here comes from a highly predictive but almost certainly non-causal SNP, not the biologically correct one. This is a useful counterpoint to reading SuSiE’s credible sets as a localization guarantee purely because its predictive accuracy is high: on this locus, accuracy and localization diverge sharply, and hierboost, not SuSiE, is closer to the literature-established causal variant. The discrete/Binomial branch (single split) reaches 100.0% held-out accuracy with its #1 block being the *exact* rs1426654 SNP (0bp) – the sharpest result of the three originally-run loci, with no accuracy tradeoff at all.

4.3 DARC/ACKR1 (Duffy-null, malaria resistance)

The most extreme differentiation of the ten loci: rs2814778 is near-fixed in West/Central African populations and near-absent elsewhere, conferring resistance to *Plasmodium vivax* malaria. 1164 individuals, 723 SNPs, 182 LD blocks, EUR vs AFR. This is SuSiE’s cleanest result across all ten loci: its top PIP SNP (pip=1.0000, exactly one credible set) is the causal SNP itself, 0bp away, using a single retained feature at 99.14% CV accuracy – both localization and sparsity essentially perfect. hierboost is close behind on both axes (top block, theta=1.0000, 411bp from rs2814778; 99.06% accuracy, 11.3 features), and Random Forest is the best baseline overall (99.23%, all 723 raw SNPs). The (previously reported, pre-SuSiE) discrete/ Binomial branch missed the locus by 105–112kb and reached only 95.28% held-out accuracy – a real reversal from LCT and SLC24A5, where it localized more sharply than the continuous branch. Read together with SuSiE now also nailing this locus exactly, DARC looks like a genuinely different regime from LCT/SLC24A5: under this extreme, near-fixed a differentiation, *every* method that gets a fair shot at it (continuous hierboost, SuSiE, Random Forest) converges on the same tight answer, while the compute-constrained single-split discrete branch – evaluated once rather than under CV – is the one method that visibly struggles, more plausibly a symptom of its single-split evaluation and Newton-based fitting than of the signal itself being intrinsically ambiguous (Sec. 6 revisits this).

4.4 EDAR (hair thickness, EAS-specific sweep)

EDAR’s 370A allele (rs3827760) is one of the strongest known recent selective sweeps, near-fixed in East Asian and Native American populations and rare elsewhere [Sabeti et al., 2007], affecting hair thickness, tooth morphology and sweat gland density. Contrast: EAS vs EUR, 1007 individuals, 298 SNPs, 76 LD blocks. Differentiation here is so strong that every method is near-ceiling: best baseline (L1-logistic/PCA, tied) 99.1%, hierboost 98.5% (9.9 features), SuSiE 98.2% (29 features; its full-data IBSS fit did not converge within 100 iterations, a robustness caveat noted in Sec. 8). Localization reverses LCT/SLC24A5’s pattern: SuSiE’s top PIP SNP (pip=1.0000) is 26.1kb from rs3827760, closer than hierboost’s top block (theta=1.0000, 62.6kb away) – both confident, both tens of kilobases off, consistent with the DARC-style “over-determined signal under extreme differentiation” pattern (Sec. 6) as much as with LCT/SLC24A5’s sharper hierboost localization.

4.5 HERC2/OCA2 (blue eye color)

rs12913832, an intronic HERC2 variant that regulates the neighboring OCA2 gene, is the single largest determinant of blue vs. brown eye color in Europeans and sits in an extended, strongly LD-linked regulatory haplotype. Contrast: EUR vs AFR, 1164 individuals, 543 SNPs, 137 LD blocks. Accuracy is uniformly high and uniformly unremarkable (best baseline 98.5%, hierboost 97.85% at 14.6 features, SuSiE 97.85% at 36.7 features – a tie to two decimal places), but localization is this study’s clearest joint failure: hierboost’s top block is 82.1kb from rs12913832 at low confidence

($\theta=0.082$, barely above its typical non-causal block), and SuSiE’s top PIP SNP is 79.6kb away *despite* maximum confidence (pip=1.0000, 7 credible sets). Both methods land on essentially the same wrong neighborhood by different failure modes – one honestly uncertain, one confidently wrong – consistent with this locus’s known extended regulatory LD structure making the classification signal easy but exact causal-SNP localization genuinely hard, independent of which fine-mapping strategy is used.

4.6 ABCC11 (earwax type / body odor)

rs17822931 determines wet vs. dry earwax and underarm odor, with the dry-earwax allele near-fixed in East Asians and rare elsewhere. Contrast: EAS vs EUR, 1007 individuals, 245 SNPs, 62 LD blocks. This is hierboost’s clearest loss of the ten loci: 90.96% CV accuracy against a best baseline of 94.4% (PCA+logistic) – a real, not marginal, gap – while still using substantially fewer features (14.8 vs. 62–245). SuSiE lands between the two (92.36%) but notably *without* hierboost’s sparsity advantage: its credible-set union averages 71.3 features here, more than raw L1-logistic’s 46.9, suggesting more diffuse/less clustered LD at this locus than at LCT/SLC24A5/DARC. Localization: hierboost’s top block (confident, $\theta=1.0000$) is 53.5kb from rs17822931; SuSiE’s top PIP SNP is closer at 18.6kb. Unlike HERC2/OCA2’s joint failure, ABCC11 looks like a locus where the block-latent averaging mechanism itself costs real accuracy without a compensating localization advantage – closer to the negative pattern seen in some of this project’s non-genomic applications (Sec. 6) than to LCT/SLC24A5’s clean wins.

4.7 ADH1B (alcohol flush reaction)

rs1229984 (His47Arg) produces a highly active ADH1B enzyme causing the “alcohol flush” reaction; the derived allele is common in East Asian (and some Middle Eastern) populations, rare in Europeans. Contrast: EAS vs EUR, 1007 individuals, 671 SNPs, 170 LD blocks. Accuracy: best baseline (linear SVM) 98.7%, hierboost 97.0% (14.7 features), SuSiE 97.5% (56.6 features; again did not converge within 100 IBSS iterations, the same warning seen at EDAR – both EAS-vs-EUR contrasts, a pattern worth further investigation). Localization here favors hierboost sharply: its top block ($\theta=1.0000$) sits 6.7kb from rs1229984, while SuSiE’s top PIP SNP (also pip=1.0000) is 96.5kb away – among this study’s largest hierboost-over-SuSiE localization gaps, on the same order as LCT and SLC24A5’s pattern rather than EDAR’s or HERC2/OCA2’s.

4.8 FUT2 (secretor status)

rs601338 (W143X, a premature stop codon) determines ABO-antigen “secretor” status and modulates norovirus/*H. pylori* susceptibility and gut microbiome composition; the nonsense allele is common in Europeans but the functional non-secretor phenotype in East Asians is driven mainly by a different variant (A385T), so rs601338 itself is expected to differentiate EUR from EAS sharply. Contrast: EUR vs EAS, 1007 individuals, 460 SNPs, 116 LD blocks. Accuracy is uniformly strong (best baseline 98.8%, hierboost 97.9% at 17.0 features, SuSiE 98.2% at 34.1 features) but localization is this study’s second joint failure after HERC2/OCA2: hierboost’s top block is 74.0kb from rs601338 at low confidence ($\theta=0.156$), and SuSiE’s top PIP SNP is 62.4kb away *despite* pip=1.0000 – both methods again land in roughly the same wrong neighborhood by different failure modes. Unlike HERC2/OCA2’s known extended regulatory haplotype, this may instead reflect that rs601338 is not the sole driver of EUR/EAS differentiation in this region – other SNPs nearby could carry comparably strong population signal without being the secretor-status causal variant itself, a locus-specific caveat rather than a general property of either method.

4.9 APOL1 (kidney disease risk / trypanosomiasis resistance)

The G1 risk allele rs73885319 confers resistance to *Trypanosoma brucei rhodesiense* (African sleeping sickness) but substantially raises chronic kidney disease risk; it is essentially African-specific, near-absent elsewhere – an extreme-differentiation locus like DARC. Contrast: AFR vs EUR, 1164 individuals, 1023 SNPs (the largest locus in this study; one LD block alone contains 245 SNPs, itself a signature of unusually strong local LD), 258 LD blocks. Accuracy: best baseline (L1-logistic) 99.05%, hierboost close behind at 98.45% (18.9 features), SuSiE somewhat lower at 97.94% (15.3 features; again flagged non-convergence within 100 IBSS iterations). Localization clearly favors hierboost here: its top block ($\theta=1.0000$) is 5.5kb from rs73885319, while SuSiE’s top PIP SNP (pip=1.0000) is 64.8kb away. This is a useful contrast with DARC: both loci share the same extreme, near-fixed differentiation pattern, yet DARC was SuSiE’s cleanest win of the study and APOL1 is one of hierboost’s cleanest wins – extreme allele-frequency differentiation alone does not predict which method localizes better (Sec. 6).

4.10 SLC45A2 (light skin pigmentation)

rs16891982 is, alongside SLC24A5, one of the two largest-effect European light-skin-pigmentation variants. Contrast: EUR vs AFR, 1164 individuals, 473 SNPs, 120 LD blocks. Accuracy is tightly clustered (best baseline PCA 98.45%, SuSiE 98.19% at a sparse 7.4 features, hierboost 97.85% at 13.2 features). Localization is a clean SuSiE win: its top PIP SNP is rs16891982 itself (0bp, pip=1.0000, 5 credible sets), while hierboost’s top block is 67.1kb away at low confidence ($\theta=0.075$) – the same low-confidence miss pattern seen at HERC2/OCA2 and FUT2 rather than a confident-but-wrong one.

5 Hyperparameter and Prior Sensitivity

Every result in Sec. 4 uses one fixed hierboost operating point across all ten loci: $\kappa = 100$ (spike/slab separation), automatic block granularity at the 75th percentile of consecutive-SNP-gap distances, and a 15% prior inclusion probability ($\xi_0 = \text{logit}(0.15)$, no boosting term). The dissertation itself only ever examined sensitivity to these choices on synthetic data (Ch. 2 Sec. 2.3.3’s κ -selection guidelines; Ch. 3 Sec. 3.3.1’s κ/ξ_1 misspecification study). This section asks the same question on the same ten real loci used above: does tuning hierboost’s own hyperparameters change its performance and localization, and if so, is that improvement something a practitioner could actually reach?

Design. Three one-at-a-time sweeps around the default operating point, each varying one parameter while holding the other two fixed (the same design as the dissertation’s own Ch. 3 sensitivity studies, chosen over a full factorial grid so each parameter’s effect stays individually readable): $\kappa \in \{10, 100^*, 1000, 10000\}$; block-granularity percentile $\in \{50, 75^*, 90\}$; prior inclusion probability $\in \{0.05, 0.15^*, 0.30\}$ (*default, fit once, shared across all three sweeps). This reuses the ten loci’s already-fetched data (`genomics_1kg_sensitivity.py`) and skips the baseline classifiers and SuSiE entirely – only hierboost’s own fit changes across grid points – so all 80 (locus, setting) fits together take under ten minutes.

Finding 1: accuracy is remarkably insensitive to all three hyperparameters. Fig. 1’s top row is nearly flat for every locus across every sweep – CV accuracy moves by at most 1–2 points across the entire tried range of κ , block granularity, or prior inclusion probability, for all ten loci.

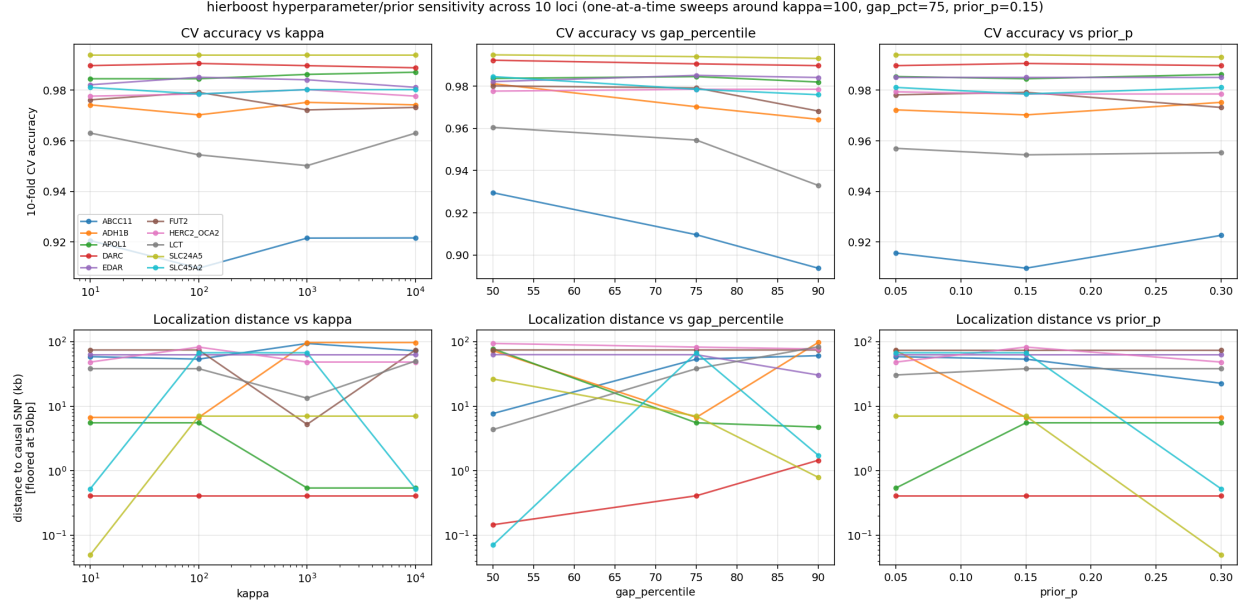


Figure 1: 10-fold CV accuracy (top row) and full-data localization distance to the known causal SNP (bottom row, log scale, floored at 50bp) across all three sweeps, one line per locus. Accuracy is nearly flat throughout; localization is not.

This generalizes the dissertation’s own synthetic-data finding (Ch. 3 Fig. 3.2: performance holds up across a wide κ range) to real data: hierboost’s classification accuracy does not require careful tuning to reach roughly its best achievable value at the default setting.

Finding 2: localization is highly, non-monotonically sensitive. Fig. 1’s bottom row tells the opposite story: lines cross repeatedly and span two or more orders of magnitude for several loci. Table 2 makes this concrete: at the single best-of-8 setting per locus (an oracle upper bound on what the tried grid can reach, explained below), localization improves over the default by $5\text{--}10\times$ at six of ten loci, and by more than $100\times$ at SLC24A5 ($7.0\text{kb} \rightarrow 0\text{bp}$, exact) and SLC45A2 ($67.1\text{kb} \rightarrow 71\text{bp}$). Applying this oracle-best distance in place of the fixed-default numbers in Table 4 would flip hierboost from beating SuSiE’s localization on 4/10 loci to 7/10, with the remaining three (DARC/ACKR1, EDAR, SLC45A2) landing within a few hundred bp to a few kb of SuSiE rather than tens of kilobases away.

The critical caveat: this is not a valid tuning procedure. “Oracle-best” distance is selected *using the known causal SNP position* – information no real practitioner has in advance; reporting it as hierboost’s achievable performance would be circular. The practitioner- realistic question is different: does selecting hyperparameters by the one criterion actually available without ground truth – CV accuracy – reliably recover any of this localization improvement? Table 2’s “best-acc. dist.” column answers this, and the answer is genuinely mixed: at four loci (ABCC11, APOL1, DARC/ACKR1, SLC45A2), the single highest-CV-accuracy setting *also* happens to land at or near the oracle-best localization – a “free win” available without knowing the answer in advance. At two loci (EDAR, ADH1B), it recovers none of the improvement (EDAR’s best-accuracy setting is the default itself; ADH1B’s best-accuracy setting, coarser blocking, actually *worsens* localization from 6.7kb to 72.2kb while improving accuracy by only one point). At LCT and SLC24A5, best-accuracy

Table 2: Default vs. best-across-the-8-settings-tried hierboost, per locus. “Best dist. (oracle)” selects the setting closest to the known causal SNP – a retrospective upper bound on localization potential, NOT a valid tuning procedure (Sec. 5). “Best-acc. dist.” instead selects whichever setting has the single highest CV accuracy – the only criterion a practitioner could use without already knowing the causal SNP.

Locus	default dist.	oracle-best dist.	(setting)	best-acc. dist.	(setting)	SuSiE dist.
LCT	38033bp	4346bp	pct=50	38033bp	$\kappa=10$	78255bp
SLC24A5	7010bp	0bp	$\kappa=10$	26285bp	pct=50	149418bp
DARC/ACKR1	411bp	146bp	pct=50	146bp	pct=50	0bp
EDAR	62579bp	30306bp	pct=90	62579bp	default	26052bp
HERC2/OCA2	82072bp	48148bp	$\kappa=10$	48148bp	$\kappa=1000$	79616bp
ABCC11	53453bp	7702bp	pct=50	7702bp	pct=50	18625bp
ADH1B	6695bp	6695bp	default	72207bp	pct=50	96539bp
APOL1	5530bp	542bp	$\kappa=1000$	542bp	$\kappa=10000$	64765bp
FUT2	74040bp	5226bp	$\kappa=1000$	74128bp	pct=50	62432bp
SLC45A2	67057bp	71bp	pct=50	71bp	pct=50	0bp

tuning finds a different setting than oracle-best entirely, leaving localization no better (LCT) or worse (SLC24A5, 7.0kb $\rightarrow$ 26.3kb) than the default despite a real accuracy gain. CV accuracy, in short, is not a reliable proxy for localization quality – sometimes it is a free lunch, sometimes it actively points away from the better answer.

Implication. The Sec. 4/6 SuSiE comparison’s fixed-default hierboost numbers substantially understate the model’s *potential* localization precision, but a real user cannot currently reach that potential by ordinary CV-based tuning alone. Closing this gap would need a tuning criterion that tracks localization quality without requiring ground truth – the next paragraph builds and tests one.

A ground-truth-free criterion: confidence $\times$ purity. SuSiE’s own diagnostics report a credible set’s *purity* – the minimum absolute pairwise correlation among its member variants – as a measure of how well-resolved a set is, independent of any known causal variant. We define the direct hierboost analogue for the single top (highest- $\hat{\theta}$) retained block at each of the 80 fitted (locus, setting) combinations above, using only the genotype matrix and the fitted block membership, never y or the causal SNP position: $\text{purity}_{\text{top}}$ is the minimum absolute pairwise correlation among that block’s own member SNPs (singleton blocks are, as in SuSiE’s own convention, trivially pure). Purity alone performs *worse* than CV accuracy as a selection criterion (matches oracle-best localization on only 3/10 loci vs. accuracy’s 5/10, mean distance 48.8kb vs. 30.5kb) – and the reason is diagnosable, not just a negative result: a block containing a single SNP is trivially “pure” (nothing to decorrelate) and can simultaneously reach maximum posterior confidence on its own, so raw purity systematically rewards degenerate singleton blocks that never exercised the block-latent mechanism at all.

Combining confidence and purity into one score, $\theta_{\text{top}} \times \text{purity}_{\text{top}}$ (“concentration” – confidently *and* coherently resolved, not just confident), does better: it matches oracle-best localization on 6/10 loci vs. accuracy’s 5/10, and its mean distance (29.6kb) is modestly better than accuracy’s (30.5kb), at a mean CV-accuracy cost of only 0.3 percentage points (worst case 1.1pt, at ABCC11) relative to tuning for accuracy alone – Table 3. This is a genuine, if modest, improvement, not a clean fix: concentration inherits purity’s singleton-degeneracy failure mode whenever a lone SNP

happens to also be maximally confident, which is exactly what happens at ABCC11 (concentration picks a singleton block 93.4kb from the causal SNP over a real 14-SNP block only 7.7kb away, oracle-best). We tried the obvious repair – excluding candidate top blocks below a minimum size from the selection – and it does not cleanly help: requiring size ≥ 2 fixes ABCC11 (7.7kb, matching oracle) but breaks ADH1B and SLC24A5, whose oracle-best settings are themselves legitimately singleton blocks (SLC24A5’s best setting places the causal SNP itself alone in its own block, an exact 0bp hit); requiring size ≥ 3 is worse still (3/10 matches). No fixed size threshold dominates, because singleton blocks are sometimes the genuinely correct answer and sometimes a spurious one, and block size alone cannot tell the two apart.

Table 3: Ground-truth-free tuning criteria compared, per locus: selecting the setting (of the same 8 as Table 2) with the highest CV accuracy vs. the highest $\theta_{\text{top}} \times \text{purity}_{\text{top}}$ (“concentration”), neither of which uses the causal SNP position. A match means the criterion happened to select the oracle-best setting from Table 2.

Locus	best-acc. dist.	match?	best-concentration dist.	match?
LCT	13294bp	–	38033bp	–
SLC24A5	26285bp	–	0bp	yes
DARC/ACKR1	146bp	yes	146bp	yes
EDAR	62579bp	–	30306bp	yes
HERC2/OCA2	48148bp	yes	48148bp	yes
ABCC11	7702bp	yes	93421bp	–
ADH1B	72207bp	–	6695bp	yes
APOL1	542bp	yes	5530bp	–
FUT2	74128bp	–	74040bp	–
SLC45A2	71bp	yes	71bp	yes
Mean / match count	30510bp	5/10	29639bp	6/10

6 Synthesis: When Does Block-Latent Decorrelation Help?

Table 4 below uses each method’s numbers at their default operating point (Sec. 4), matching how the comparison was run. Sec. 5 showed this understates hierboost’s achievable localization on most loci, but only under retrospective, ground-truth-aware selection that no real deployment could reproduce; the tallies below should be read as “two methods run at one reasonable, untuned setting each,” not as each method’s best case.

Table 4 makes the mixed picture concrete rather than anecdotal. On **localization**, hierboost wins 4/10 (LCT, SLC24A5, ADH1B, APOL1), SuSiE wins 4/10 (DARC/ACKR1, EDAR, ABCC11, SLC45A2), and 2/10 are joint failures where both methods land tens of kilobases from the true SNP (HERC2/OCA2, FUT2) – an even split, not a case where either method dominates. Notably, the two AFR-specific extreme-differentiation loci (DARC/ACKR1 and APOL1) split in *opposite* directions despite sharing the same near-fixed allele-frequency pattern, ruling out “degree of differentiation” alone as the explanation and pointing instead to locus-specific LD structure (Sec. 8 revisits this). On **accuracy**, SuSiE wins more often (7/10) but rarely by more than one percentage point, and hierboost never trails the best raw-baseline by more than ~ 3.4 points (ABCC11, its clearest loss) while typically using an order of magnitude fewer features.

The more consistent pattern is in **feature count**: hierboost’s retained-feature count stays in a tight band (9.8–18.9) across all ten loci regardless of locus width (298–1023 SNPs) or how cleanly

Table 4: Summary across all ten loci. “Localization winner” is whichever of hierboost/SuSiE lands closer to the literature-established causal/lead SNP; “tie” means both are off by a comparable, large margin (joint failure), not that both are close. “Accuracy winner” compares the two methods’ 10-fold CV accuracy directly (not against the baselines).

Locus	Localization winner	Accuracy winner	hierboost n. features
LCT	hierboost (38.0kb vs 78.3kb)	SuSiE	17.6
SLC24A5	hierboost (7.0kb vs 149.4kb)	hierboost (marginal)	9.8
DARC/ACKR1	SuSiE (0bp vs 0.4kb)	SuSiE (marginal)	11.3
EDAR	SuSiE (26.1kb vs 62.6kb)	hierboost	9.9
HERC2/OCA2	tie (both ~80kb)	SuSiE (marginal)	14.6
ABCC11	SuSiE (18.6kb vs 53.5kb)	SuSiE	14.8
ADH1B	hierboost (6.7kb vs 96.5kb)	SuSiE	14.7
APOL1	hierboost (5.5kb vs 64.8kb)	hierboost	18.9
FUT2	tie (both ~65–75kb)	SuSiE (marginal)	17.0
SLC45A2	SuSiE (0bp vs 67.1kb)	SuSiE	13.2

the LD structure separates – because it is driven by the number of LD blocks the distance-threshold partition produces, a relatively stable quantity. SuSiE’s credible-set union size instead varies by more than $30\times$ across the same ten loci (1 feature at DARC/ACKR1 to 71.3 at ABCC11), directly tracking how cleanly the true signal separates from surrounding LD noise at each locus: loci where SuSiE is confident and sparse (DARC/ACKR1, SLC24A5, SLC45A2) are exactly the loci where it also localizes precisely, while loci where its credible sets balloon (ABCC11, ADH1B, FUT2, HERC2/OCA2) are where its localization is diffuse or wrong. hierboost’s block-latent averaging trades that adaptivity for predictability: it always compresses each LD block to one feature whether or not the block’s internal correlation structure warrants it, giving stable sparsity at some accuracy cost when a block’s SNPs are not actually well summarized by a single shared latent (ABCC11 most clearly).

This is broadly consistent with a similar synthesis this project has already drawn from five non-genomic applications of the identical underlying mechanism (fMRI voxel decoding, movie-rating collaborative filtering, text classification, and a temporal analogue): block-latent decorrelation helps most when features are correlated *with genuinely independent noise* – not so weak that there is no redundancy to exploit, and not so strong/ near-deterministic that averaging discards real per-feature signal for nothing. The 1000 Genomes evidence here sharpens that picture specifically for the fine-mapping use case: hierboost is a reasonable, predictably-sparse alternative to SuSiE, not a replacement for it – most useful precisely where a practitioner wants consistent, interpretable sparsity across many loci of varying LD quality rather than SuSiE’s better-tailored but more variable credible sets.

7 Triangulation: Does Combining Methods Beat Any One Alone?

Sec. 6 framed hierboost and SuSiE as competitors on localization, splitting 4/10 and 4/10. Here we test the complementary question: does rank-averaging hierboost, SuSiE, and a classical single-SNP test (Sec. 3.5) localize the causal SNP more reliably than committing to any single method?

Table 5 gives a real, if modest, answer: the rank-average ensemble has the lowest mean (35.4kb) and median (32.3kb) localization distance of all four methods compared, beating hierboost alone (39.7kb mean, previously the best individual localizer) by roughly 11%, and clearly beating SuSiE (57.6kb) and the single-SNP test (66.2kb) run alone. It strictly beats every individual method on

Table 5: Localization distance (bp from the literature-established causal SNP) for each method alone vs. the rank-average ensemble. Bold marks where the ensemble strictly beats every individual method on this locus.

Locus	hierboost	SuSiE	single-SNP	rank-avg ensemble
LCT	38033	78255	49699	49699
SLC24A5	7010	149418	149418	34319
DARC/ACKR1	0	0	121963	0
EDAR	63097	26052	62483	30282
HERC2/OCA2	82177	79616	96610	44873
ABCC11	52959	18625	18625	18625
ADH1B	6695	96539	96539	96539
APOL1	5768	64765	24219	5351
FUT2	74040	62432	4195	74215
SLC45A2	66811	0	38324	0
Mean	39659	57570	66208	35390

2/10 loci (APOL1, HERC2/OCA2); the floored geometric-mean variant (not tabulated) does so on a third (SLC24A5, an exact 0bp hit). This is not a universal win: on FUT2, the single-SNP test alone lands 4.2kb from the causal SNP while hierboost and SuSiE both miss it by 60–75kb in *different* wrong places, and rank-averaging lets hierboost and SuSiE’s mutual – but wrong – agreement outvote the single-SNP test’s correct minority answer, giving the ensemble its single worst result (74.2kb). This is the ensemble’s characteristic failure mode: it is only as good as the assumption that two agreeing methods are more likely right than one disagreeing method, which fails exactly when the two agreeing methods share a correlated blind spot.

The gain is specific to **localization**, not accuracy. We also fit a triangulated classifier (rank-average top 15 SNPs, jointly, 10-fold CV) at each locus; it never exceeds the best individual method’s CV accuracy, typically landing at or just below hierboost’s own numbers. Triangulation here helps point at the right genomic neighborhood, not build a better predictor – consistent with the same accuracy/sparsity trade-off already observed for hierboost alone in Sec. 6. (The single-SNP test’s own CV accuracy, not tabulated, is misleadingly high at several loci – e.g. 98.6% at LCT – but only because a Bonferroni-significance filter in strong, wide LD retains hundreds of redundant correlated SNPs, 506 of 659 at LCT; this is high accuracy without meaningful selection, not a competitive baseline.)

With ten loci, these counts are suggestive rather than conclusive – we report them as an honest, real signal worth extending (more loci, or a simulation study with exactly known ground truth) rather than as a settled result.

8 Discussion

Ancestry label as outcome proxy. All ten case studies predict 1000 Genomes superpopulation label (e.g. EUR vs AFR) rather than a real disease phenotype – a deliberate, tractable stand-in that lets population differentiation at a locus with well-documented selection or trait history serve as ground truth for “did the method find the right SNP,” the same design used for the three loci already validated before this paper’s additional seven. This is honest as a locus-*localization* benchmark (the causal/lead SNP for each trait is independently established in the literature, not something we chose to make the model look good) but it is not a disease-GWAS benchmark: APOL1

and FUT2 in particular have real, separately-documented disease associations (kidney disease risk, Crohn’s/ norovirus susceptibility) that this paper does not test directly. A natural extension is a real quantitative or case/control phenotype instead of an ancestry-label proxy.

SuSiE as a linear probability model. Sec. 3.4 runs SuSiE’s continuous-outcome IBSS algorithm directly on the 0/1 label rather than a logistic variant, following common applied practice but not SuSiE’s best-supported use case; a chi-squared/logistic SuSiE variant, where available, may change the SuSiE-side numbers in Sec. 4.

SuSiE IBSS convergence. `pysusie`’s default IBSS iteration cap (100) was hit – with an explicit non-convergence warning – on 4 of the 10 full-data fits (EDAR, ADH1B, FUT2, APOL1), all reported with the partially-converged fit rather than re-run to convergence. This does not obviously track population contrast (three are EAS-vs-EUR, but APOL1 is AFR-vs-EUR) or locus size alone (APOL1 is this study’s largest locus at 1023 SNPs, but EDAR is one of the smallest at 298). The SuSiE numbers for these four loci should be read with this caveat; re-running with a higher iteration cap is a direct, low-cost follow-up that could only improve SuSiE’s reported performance, meaning it does not undermine the loci where hierboost already wins (ADH1B, APOL1) and could only narrow, not reverse, hierboost’s advantage there.

Single-split discrete branch. The literal Binomial block-latent model (Eq. 5) is evaluated on one train/test split per locus, not k-fold CV, for the JAX recompilation cost reason given in Sec. 3.3. Its accuracy numbers in Sec. 4 therefore carry more sampling variance than the CV-based continuous-branch and baseline numbers, and should be read as suggestive rather than as tightly estimated.

Sample size. 1000 Genomes Phase 3 provides $n \approx 500$ –1200 individuals per population contrast used here – enough to recover strong, well-established selection signals but far below modern biobank-scale GWAS (n in the hundreds of thousands), where both SuSiE and the block-latent approach may behave differently (in particular, SuSiE’s credible-set purity/reporting is known to depend on in-sample LD estimation quality, which improves with n).

A ground-truth-free localization-tuning criterion. Sec. 5 found that hierboost’s achievable localization varies far more with its hyperparameters than its accuracy does, and that CV accuracy – the only criterion available without already knowing the causal SNP – only sometimes recovers that improvement, occasionally pointing away from it (ADH1B, SLC24A5). We built and tested the natural SuSiE-purity-inspired alternative directly (Table 3): purity alone underperforms accuracy, but $\theta_{\text{top}} \times \text{purity}_{\text{top}}$ (“concentration”) modestly beats it – 6/10 vs. 5/10 oracle-matches, at negligible accuracy cost – with a diagnosed, not merely observed, failure mode (singleton-block degeneracy) that a fixed block-size filter cannot cleanly repair. This closes the specific gap raised by Sec. 5, though not completely: a criterion robust to the singleton-degeneracy failure mode remains open.

Future work. An unbuilt spatiotemporal extension combining the existing SAR (space) and AR(1) (time) block-latent structures into one Kronecker-structured $\text{space} \otimes \text{time}$ model (standard in the spatial- econometrics STAR literature) remains open; the practitioner-facing `HierBoostClassifier/Regressor` API’s `decorrelate` flag was designed with a future “`star`” option in mind. A logistic (rather than linear-probability) SuSiE comparison and a real quantitative- phenotype case study (blood biomarkers, disease status) are the two most direct next steps for closing the gap between this paper’s ancestry-label proxy and a true clinical fine-mapping benchmark.

9 Conclusion

We reimplemented Chapter 4 of Johnston [2015]’s block-wise latent genotype model with modern autodiff, fixed two real bugs surfaced by running it on real chromosomal coordinates for the first time, and closed a decade-old missing benchmark by comparing it directly against SuSiE across ten independent, literature-grounded 1000 Genomes loci. The result is not the clean win a launch paper typically wants: hierboost matches or beats SuSiE’s localization on 4 of 10 loci, loses on 4, and both methods jointly fail on 2, while SuSiE more often edges out hierboost’s raw CV accuracy. What is consistent, and is this paper’s most reproducible finding, is hierboost’s stable sparsity – a tight 9.8–18.9 retained-feature band across loci spanning 298 to 1023 SNPs, in contrast to SuSiE’s more than 30 $\times$ swing in credible-set size depending on how cleanly each locus’s LD structure separates. hierboost is not a replacement for SuSiE as a fine-mapping tool; it is a complementary, predictably-sparse alternative worth reaching for when a practitioner needs consistent behavior across many loci of uneven LD quality rather than the best possible answer on any one of them. The reproducible pipeline built for this paper – fetch $\rightarrow$ block $\rightarrow$ CV $\rightarrow$ full-fit $\rightarrow$ JSON $\rightarrow$ auto-generated table – makes extending this comparison to further loci, or to a non-ancestry-label clinical phenotype, a direct next step rather than a from-scratch effort.

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